SUPERKEY AND CANDIDATEKEY

K KRISHNA PRIYA

CSE DEPT

SUPERKEY

- Any set of attributes of a table that can uniquely identify all the tuples of that table is known as a **Super key**.
- It's different from the primary and candidate keys in the sense that only the minimal superkeys are the candidate/primary keys.
- This means that from a super key when we **remove** all the attributes that are **unnecessary** for its uniqueness, only then it **becomes a primary/candidate** key. So, in essence, all primary/candidate keys are super keys but not all super keys are primary/candidate keys.
- By the formal definition of a Relation(Table), we know that the tuples of a relation are all unique. So, the set of all attributes itself is a super key.

NUMBER OF POSSIBLE SUPER KEYS IN DBMS

- Let a Relation R have attributes {a1, a2, a3,…,an}. Find Super key of R.
- **Maximum Super keys = 2^n – 1**.
- **Proof:** There are n attributes in R. So the total number of possible subsets/combination of attributes of R is 2^n
- Now to be a Super key, there should be **at least one** attribute present i.e. the NULL set or the set with no attribute can't be a super key.
- So, maximum possible number of Super keys of R = 2^n – 1.

FINDING SUPERKEYS WITH CANDIDATE KEY

- Let a Relation R have attributes {a1,a2,a3} and a1 is the candidate key. Then how many super keys are possible?
- Here, any superset of a1 is the super key.

  Super keys are = {a1, a1 a2, a1 a3, a1 a2 a3}

  Thus we see that 4 Super keys are possible in this case.
- In general, if we have 'N' attributes with one candidate key then the number of **possible superkeys is $2^{(N-1)}$.**

EXAMPLE-1:

- Let a Relation R have attributes {a1, a2, a3,…, an} and the candidate key is “**a1 a2 a3**” then the possible number of super keys?
- Following the previous formula, we have 3 attributes instead of one. So, here the number of possible super keys is $2^{(N-3)}$.

EXAMPLE-2:

- Let a Relation R have attributes {a1, a2, a3,…, an} and the candidate keys are "a1", "a2" then the possible number of super keys?
- This problem now is slightly different since we now have two different candidate keys instead of only one. Tackling problems like these is shown in the diagram below:

= (super keys possible with candidate key A1) + (super keys possible with candidate key A2) – (common superkeys from both A1 and A2)
= 2(n-1) + 2(n-1) - 2(n-2)

|A1 ∪ A2| = |A1| + |A2| - |A1 ∩ A2|

- Example3 -: Let a Relation R have attributes {a1, a2, a3,…, an} and the candidate keys are “a1”, “a2 a3” then the possible number of super keys?
- Super keys of (a1) + Super keys of (a2 a3) – Super keys of (a1 a2 a3)
- $\Rightarrow 2^{(n-1)} + 2^{(n-2)} - 2^{(n-3)}$

- **Example-4:** Let a Relation R have attributes {a1, a2, a3,…, an} and the candidate keys are “a1 a2”, “a3 a4” then the possible number of super keys?
- Super keys of(a1 a2) + Super keys of(a3 a4) – Super keys of(a1 a2 a3 a4)
- $\Rightarrow 2^{(n-2)} + 2^{(n-2)} - 2^{(n-4)}$

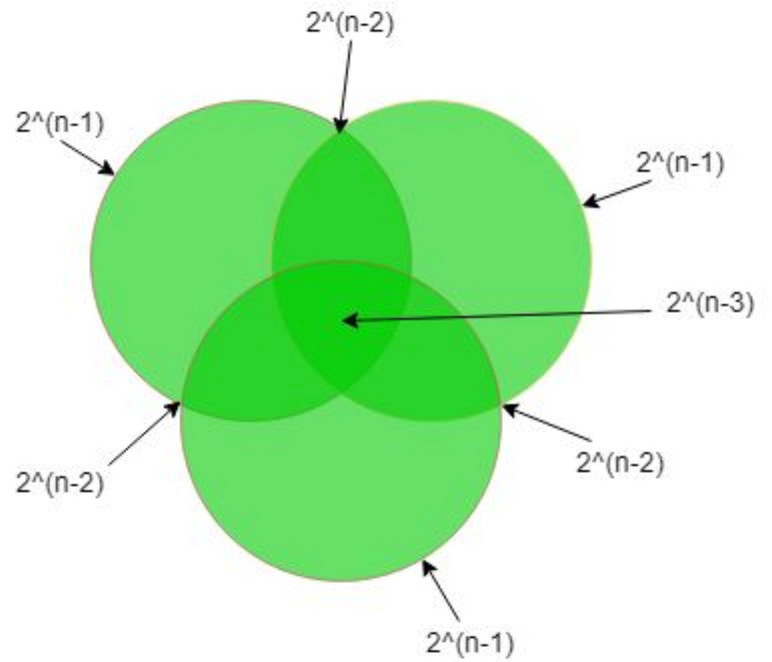
EXAMPLE-5

- Let a Relation R have attributes {a1, a2, a3,…, an} and the candidate keys are "a1 a2", "a1 a3" then the possible number of super keys?
- Super keys of (a1 a2) + Super keys of (a1 a3) – Super keys of(a1 a2 a3)
- $\Rightarrow 2^{(n-2)} + 2^{(n-2)} - 2^{(n-3)}$
- Note: Number of super keys = 2^{n-k} where "n" is the number of attributes in the Relation R and "k" is the number of attributes in the Candidate Key

EXAMPLE-6 :

- Let a Relation R have attributes {a1, a2, a3,…,an} and the candidate keys are “a1”, “a2”, “a3” then the possible number of super keys?

- In this question, we have 3 different candidate keys. Tackling problems like these are shown in the diagram below.



|A1 ∪ A2 ∪ A3| = |A1| + |A2| + |A3| – |A1 ∩ A2| – |A1 ∩ A3| – |A2 ∩ A3| + |A1 ∩ A2 ∩ A3|
= (super keys possible with candidate key A1) + (super keys possible with candidate key A2) + (super keys possible with candidate key A3) – (common super keys from both A1 and A2) – (common super keys from both A1 and A3) – (common super keys from both A2 and A3) + (common super keys from both A1, A2, and A3)

$$\mathbf{2^{(n-1)} + 2^{(n-1)} + 2^{(n-1)} - 2^{(n-2)} - 2^{(n-2)} - 2^{(n-2)} + 2^{(n-3)}}$$

- Let a relation R have attributes {a1,a2,a3_______ an} such that any k of the attributes at a time determines all other attributes. Find the value of k such that the number of candidate keys in the relation will be maximum.
- Any k attributes at a time constitute one candidate key. These k attributes are randomly chosen from the n attributes.
- So for some k, the possible no of candidate keys is n_{C_k},i.e,
- n!/(n-k)!k! . For the number of members to be maximum k must be ⌊n/2⌋ so that nCk is the maximum for that value.

- **Example-7:** A relation R (A, B, C, D, E, F, G, H)and set of functional dependencies are {$CH \rightarrow G$, $A \rightarrow BC$, $B \rightarrow CFH$, $E \rightarrow A$, $F \rightarrow EG$ }
- Then how many possible super keys are present?

- Step 1:- First of all, we have to find what the candidate keys are:-
- as we can see in the given functional dependency D is missing but in relation, D is given so D must be a prime attribute of the Candidate key.
- $A+ = E+ = B+ = F+ =$ all attributes of a relation except D
- So, Candidate keys are = AD, BD, ED, FD
- **Step 2:-**Find super keys due to a single candidate key there is a two possibilities of attribute either we select or not hence there will be 2 chances so,

TO FIND THE SUPER KEY OR CANDIDATE KEY, WE HAVE TWO METHODS:

- Given Relation:
- **R (A, B, C, D, E, F)**
- Given Functional Dependencies (F):{ AB → C ,BC → AD, D → E, CF→ B}
- The first method is to get the Closure of Attributes for each attribute and the possible combinations. And then figure out which attribute(s) can determine all other attributes. This process is too lengthy.
- Thus we will use the shortcut. The shortcut is to check for the attribute(s) which are NOT present on the Right Side of any Functional Dependency.
- The attribute F is not present on the Right Side of any Functional Dependency. This means that the attribute F cannot be determined by any other attribute(s). Thus F must be present on the candidate key and also the super key.

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So let's start with finding the closure of F.
F+ = {F}
From the given set of functional dependencies, the attribute F cannot determine any other attribute. Thus F canNOT be the Super Key.

Let's find the closure of the attribute CF.
{CF} +:= {CF} (because of the axiom of reflexivity)
= {CFB} (from the given dependency CF $\rightarrow$ B)
= {CFBAD} (from the given dependency BC $\rightarrow$ AD)
= {CFBADE} (from the given dependency D $\rightarrow$ E)

Finally {CF}+ = {CFBADE} or {ABCDEF}

- Since CF can determine all other attributes, CF is a Candidate Key.
- And all the combinations of CF with or without other attribute(s) are Super Keys.
- For example, CF, CFA, CFB, CFD, CFE, ABCDF, ABCFE, etc are Super Keys.
- We have already determined one Candidate Key which is CF. To check if any other Candidate Key exists, let's see if C or F is replaceable by any other attribute(s). As we have a dependency
- $AB \rightarrow C$,
- we can replace C with AB. Thus now we have: The Closure of ABF:
- $\{ABF\}^+ = \{ABCDEF\}$ So now, Candidate Keys = {CF, ABF}
- Now we can say:
- **Prime Attributes** = {A, B, C, F}, attributes which are part of Candidate Keys.

<u>FINDING CANDIDATE KEYS-</u>

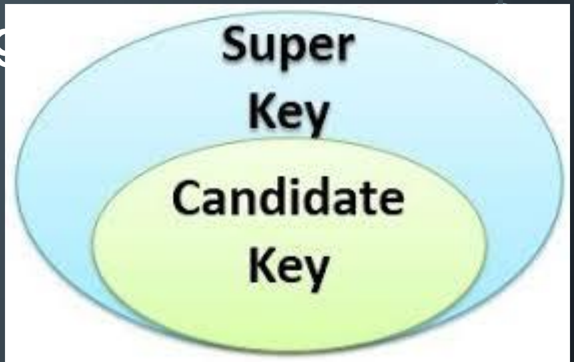
- We can determine the candidate keys of a given relation using the following steps-
- **<u>Step-01:</u>**
-
- Determine all essential attributes of the given relation.
- Essential attributes are those attributes which are not present on RHS of any functional dependency.
- Essential attributes are always a part of every candidate key.
- This is because they can not be determined by other attributes.

- **<u>Example</u>**
- Let R(A, B, C, D, E, F) be a relation scheme with the following functional dependencies-
- $A \rightarrow B$
- $C \rightarrow D$
- $D \rightarrow E$
- Here, the attributes which are not present on RHS of any functional dependency are A, C and F.
- So, essential attributes are- **A, C and F**.

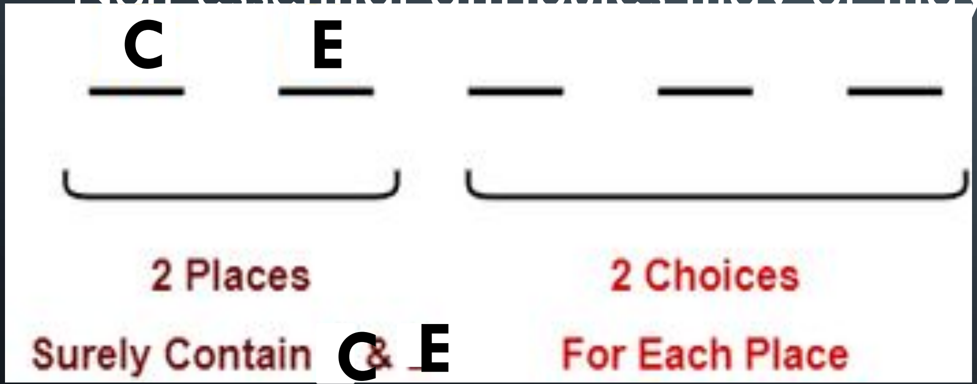
- **Step-02:**
- The remaining attributes of the relation are non-essential attributes.
- This is because they can be determined by using essential attributes.
- Now, following two cases are possible-
- **Case-01:**
- If all essential attributes together can determine all remaining non-essential attributes, then-
- The combination of essential attributes is the candidate key.
- It is the only possible candidate key.
- **Case-02:**
- If all essential attributes together can not determine all remaining non-essential attributes, then-
- The set of essential attributes and some non-essential attributes will be the candidate key(s).
- In this case, multiple candidate keys are possible.
- To find the candidate keys, we check different combinations of essential and non-essential attributes.

- Let R = (A, B, C, D, E, F) be a relation scheme with the following dependencies-
- C → F ,E → A, EC → D, A → B
- Which of the following is a key for R? Also, determine the total number of candidate keys and super keys.
- **Step-02:**
- Now, We will check if the essential attributes together can determine all remaining non-essential attributes. To check, we find the closure of CE. So, we have- $\{ CE \}^+$
- = { C , E }
- = { C , E , F }   (Using C → F)
- = { A , C , E , F }   (Using E → A)
- = { A , C , D , E , F }   (Using EC → D)
- = { A , B , C , D , E , F }   (Using A → B)
- We conclude that CE can determine all the attributes of the given relation.

- **Total Number of Candidate Keys-** Only one candidate key CE is possible.
- **Total Number of Super Keys-** There are total 6 attributes in the g



  of which- There are 2 essential attributes- C and E.
- Remaining 4 attributes are non-essential attributes.
- Essential attributes will be definitely present in every key.
- Non-essential attributes may or may not be taken in every super key



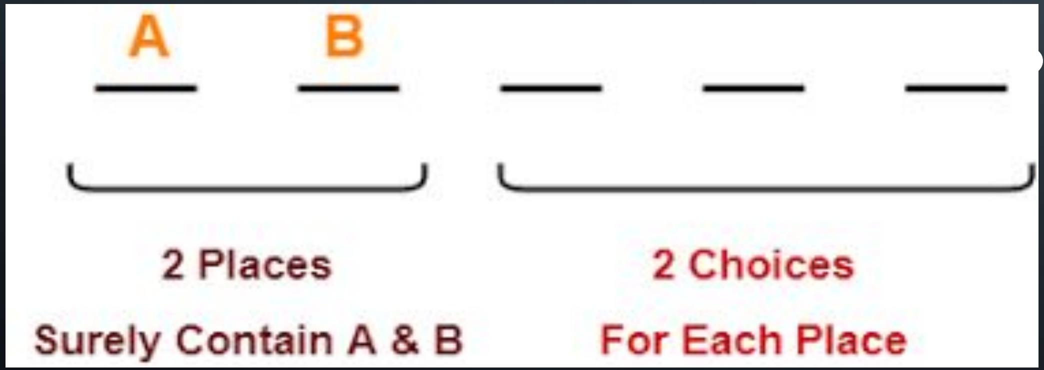
So, number of super keys possible = 2 x 2 x 2 x 2 = 16.
Thus, total number of super keys possible = 16.

EXAMPLE 2: LET R = (A, B, C, D, E) BE A RELATION SCHEME WITH THE FOLLOWING DEPENDENCIES- DETERMINE THE TOTAL NUMBER OF CANDIDATE KEYS AND SUPER KEYS.

- $AB \rightarrow C$,$C \rightarrow D$,$B \rightarrow E$
- **<u>Solution-</u>**We will find candidate keys of the given relation in the following steps-
- **<u>Step-01:</u>** Determine all essential attributes of the given relation.
- Essential attributes of the relation are- A and B.
- So, attributes A and B will definitely be a part of every candidate key.

- **<u>Step-02:</u>** Now,
- We will check if the essential attributes together can determine all remaining non-essential attributes.
- To check, we find the closure of AB.
- So, we have- $\{ AB \}^+$
- $= \{ A , B \}$
- $= \{ A , B , C \}$ (Using $AB \rightarrow C$)
- $= \{ A , B , C , D \}$ (Using $C \rightarrow D$)
- $= \{ A , B , C , D , E \}$ (Using $B \rightarrow E$)
- We conclude that AB can determine all the attributes of the given relation.
- ***Thus, AB is the only possible candidate key of the relation.***

- **<u>Total Number of Candidate Keys-</u>**
- Only one candidate key AB is possible.
- **<u>Total Number of Super Keys-</u>**
- There are total 5 attributes in the given relation of which-
- There are 2 essential attributes- A and B.
- Remaining 3 attributes are non-essential attributes.
- Essential attributes will be definitely present in every key.



t be taken in every super key.

So, number of super keys possible = 2 x 2 x 2 = 8.

Thus, total number of super keys possible = 8.

- **Problem** Consider the relation scheme R(E, F, G, H, I, J, K, L, M, N) and the set of functional dependencies-
- { E, F } → { G }
- { F } → { I , J }
- { E, H } → { K, L }
- { K } → { M }
- { L } → { N }
- Also, determine the total number of candidate keys and super keys.
- If a combination of essential attributes is not a candidate key, then we will try different combinations of essential and non-essential attributes which are not

- Example 1:
- Let's suppose we have a set of attributes as S: {A, B, C, D} and functional dependencies are:
- A -> B ,B -> C, C -> A
- **Solution:**
- In the above example, we have non-essential attributes as {A, B, C}, and the essential attribute is {D}.
- So, D will be the part of the candidate key.
- Closure of D: {D}
- So, a combination of essential attributes is not able to find all the attributes, so we will add non-essential attributes in different ways to find out the candidate keys.
- Closure of (AD) = {A, B, C, D} (using A->B and B->C)
- So, CD will be a candidate key.

- **Example 2:**
- Let's suppose we have a set of attributes as S: {A, B, C, D, E, F} and functional dependencies are:
- AB -> C
- B -> AE
- C -> D
- **Solution:**
- In the above example, we have non-essential attributes as {A, C, D, E}, and the essential attribute is {B, F}.
- So, BF will be the part of the candidate key.
- Closure of BF: {B, A, E, C, D, F}
- Since the combination of essential attributes is capable of finding all the attributes, it will be a candidate key. All other combinations of the essential and non-essential